CEE 181 and 281 Design Project Guide

Fall 2026 | A continuing design project

Develop a common freeway simulation and traffic management foundation, then complete ONE required final control component: CAV control or traffic signal control. The shared foundation includes observation, estimation, prediction and a basic closed-loop TMC. This is one continuing project with three intermediate submissions and a final presentation.

A common environment with interchangeable components

The shared environment is a bounded freeway segment with arrivals, and a bottleneck - a controlled ramp-entry queue. After the basic TMC works, the CAV pathway extends vehicle control in this environment; the signal pathway adapts reusable vehicle/queue and observation modules to a supplied isolated intersection. The signal pathway implements actual signal phases and stop-line service, not just freeway ramp metering.

The simulation supplies states and virtual detectors. Observation components produce estimates and forecasts and include a short video exercise. The basic TMC uses permitted measurements to make decisions. Simulator truth is withheld for evaluation. In the final pathway, students reuse these interfaces while changing the controlled traffic behavior or signal phase.

Component	Minimum exchange format (Example)
Vehicle state	time_s, vehicle_id, lane_id, position_m, speed_mps; optional acceleration_mps2. Define origin, direction and interval.
Observation	sensor_id, interval_start_s, interval_end_s, count, mean_speed_mps when measurable, quality_flag. Unavailable quantities remain explicitly missing.
Estimate and forecast	issue_time_s, target_time_s, segment_id, estimated_state or predicted_value, variable_name, units, quality or uncertainty indicator.
Control command	issue_time_s, apply_time_s, actuator_id, action, value, units. Include valid bounds and fallback state.
Evaluation record	scenario_id, demand_id, seed when stochastic, model_version, controller_version and metrics including boundary queues.

Interface rules

Use seconds and meters internally, explicit sampling intervals and consistent vehicle identifiers. Convert units at documented boundaries. Agree an initial interface before component exchange. Each module must run with a small example.

Implementation workload

Target approximately five hours per student per week outside class. Full lane changing, a large network, detector training from scratch and reinforcement learning are optional extensions. Complete one final pathway; neither completing both nor adding software complexity earns automatic credit.

Simulation component and behavioral investigation (Draft)

Question: Under what conditions does a traffic disturbance disappear, grow or produce a persistent queue, and how do model assumptions change the answer?

Scope and literature

Implement vehicle movement in Python, or any programming language, using a justified car-following model such as IDM. Add arrivals, exits, a bottleneck and virtual detector outputs. Read at least two substantive sources, including one original model source. Connect specific equations, parameter ranges and limitations to the sources. Suggested course reference: Traffic Flow Dynamics, selected sections of Chapters 2–4, 10 and 11.

Design task	Evidence to develop
Model design	Select and justify a model, equations, units, parameters and limitations. Build a runnable minimum case and predict a response before testing it.
Verification	Check conservation, free motion or equilibrium, gaps, boundary behavior and time-step sensitivity. Audit detector counting.
Investigation	Test a focused claim using a baseline and meaningful changed conditions. Show trajectory and aggregate plots and an independent analytical or literature comparison.

Useful investigations

- Does a lead-vehicle braking disturbance grow or decay for the selected headway and demand conditions?
- How do driver heterogeneity and bottleneck capacity affect queue formation and discharge?
- Does a conclusion survive a smaller time step or a different initial condition? Explain disagreements rather than hiding them.

Roles and individual questions

Example pair roles: A owns vehicle dynamics and numerical verification; B owns arrivals, bottleneck logic and detector measurements. Both trace a vehicle through the system and explain a test of the other component's interface. Questions: Where is the position updated? Why is this parameter plausible? How can we distinguish a queue from a coding artifact?

Submission

Include the simulator, checks and behavioral results in intermediate submission 1. Maintain one repository and one evolving design record. CEE281 adds a justified comparison with a second model or independent calibration/validation evidence. Calibration requires suitable data; parameter tuning alone is not validation.

Observation estimation and prediction component

Question: What traffic state can we infer from limited measurements, and how accurately can we predict what happens next?

Scope and literature

Develop an observation pipeline that accepts the simulator's detector outputs. Count crossings or estimate a queue and audit a labeled sample manually. Existing detector/tracker software may be used with attribution.

Use at least two relevant sources addressing measurement, estimation or prediction. Implement a state estimate and one short-horizon forecast. Use a simple baseline such as a moving average. The core forecast can be queue length or interval flow, with a declared horizon such as 30 or 60 seconds. Suggested course reference: Traffic Flow Dynamics, selected sections of Chapters 3, 5 and 16.

Design task	Evidence to develop
Observation	Build sensors and collect data.
Estimation and forecasting	Define the state and forecast horizon, compare against a simple baseline, separate development and evaluation periods.

Evaluation that tests the actual claim

Use an appropriate error measure with its units, such as count error per interval or queue-length MAE in vehicles. Compare forecasting methods at identical horizons and on identical evaluation periods. Test at least one observation defect: missing intervals, delayed reports, noise or reduced detector coverage. Evaluate simulation estimates against withheld truth.

Roles and individual questions

Example pair roles: A owns measurement and the observation adapter; B owns estimation and forecasting. Both run the other component on a shared test case. Questions: What can this sensor actually measure? Which time period fitted this parameter? Why does your forecast beat baseline? Can this estimator run on a peer's traffic without changing its internal logic?

Submission

Include observation, estimation, forecasting and component-exchange evidence in intermediate submission 2.

Common TMC foundation and transition to final control

Question: Can the shared system turn limited observations into a bounded action and measure the subsequent traffic response?

Scope and literature

Integrate simulation, observations, estimation, prediction and a simple ramp-release decision. Implement a minimal feedback rule and compare it with a no-control or fixed-control case. This checks that the loop works before the final CAV or signal extension. Common ramp metering is a foundation exercise.

Each update follows observe, estimate, predict, decide, act and evaluate. Log the available observations, estimate, forecast, action and response. A forecast may inform the action or be an advisory output; disclose its role. Read the course control material and at least one relevant technical source before implementing the final controller.

Design task	Evidence to develop
Basic operational design	Define observations, bounds, a simple ramp-release rule, storage constraints and a fallback. This is the shared foundation.
Working loop	Trace observation, estimate, decision, action and response. Compare one simple baseline and a sophisticated metering algorithm.
Transition to final pathway (next two pages)	Demonstrate the basic TMC and an initial CAV or signal prototype. Reuse the same logging and observation interfaces.

Fair comparisons and bounded claims

Use consistent initialization, demand, simulation duration and measurement definitions. Include residual queues or run long enough to clear them. Report unmet demand rather than silently dropping vehicles at a full boundary. A deterministic model does not need arbitrary repeated identical runs. Operational improvements do not establish crash reduction; surrogate measures require explicit definitions, suitable trajectories and qualified interpretation.

Roles and individual questions

Example pair roles: A owns control logic; B owns integration, comparative experiments and performance accounting. Each student independently explains a decision trace and runs a changed scenario. Questions: What did the controller know at this instant? Which action would change with a different estimate? Does the result improve when all waiting vehicles are counted?

Submission

The final submission must include the selected CAV or signal implementation, not only this common TMC. Show baseline comparisons, a changed-demand test, a failure or degraded-information test and an individual design defense. CEE281 adds a focused controller comparison, uncertainty analysis or ablation. A supported negative result can earn full credit.

Final pathway A Connected and automated vehicle control

Choose this pathway OR the traffic-signal pathway. Implement a vehicle-level control mechanism and show how it changes traffic behavior in the shared simulation. A discussion of CAV benefits or a generic change in car-following parameters alone is insufficient.

A bounded design question

Example: Can an ACC or cooperative ACC controller reduce the growth of braking disturbances while preserving throughput and acceptable gaps in mixed traffic? Alternatively, implement a connected speed-advisory strategy with an explicit driver-compliance model. Select one mechanism and an operational question before expanding the scope.

Required implementation

- Define the control law, measured inputs, update interval, acceleration/braking bounds and fallback. For cooperative control, specify exactly which communicated vehicle states are used and how delayed or missing messages are handled.
- Implement human-driven and equipped vehicles, a declared equipment-penetration setting, and an interface between observations and commands. Advice requires modeled compliance; automated control requires a defined command-to-motion response.
- Keep future trajectories and simulator truth out of the controller inputs. Use truth only to score behavior. Check time-step sensitivity and boundary cases before interpreting a gain.

Minimum evaluation

- Compare with a credible human-driven baseline under the same demand, initial conditions and lead-vehicle disturbance. For a cooperative design, add a noncooperative ACC comparison to isolate the value of communication.
- Test nominal conditions, at least two nonzero penetration settings in addition to the unequipped baseline, and a delay/dropout or noncompliance condition relevant to the design. Include changed demand or a stronger disturbance.
- Report throughput and total delay, plus trajectories, disturbance amplification and minimum gaps or braking behavior. Explain tradeoffs; these simulation indicators do not establish real-world crash reduction.

Literature roles and presentation

Use at least two substantive sources for the final pathway, including an original controller paper or authoritative technical description. Trace equations and parameter choices to their sources. One partner may own control and communication/compliance logic; the other owns integration and comparative experiments. Each student must explain a decision trace and predict a changed-case response.

In the final presentation, show the baseline and controlled case, explain the mechanism, identify a failure condition and defend whether the evidence supports adoption. CEE281 adds a controller comparison, uncertainty study or ablation that isolates an important design choice.

Final pathway B Traffic signal control

Choose this pathway OR the CAV pathway. Reuse the common simulation, observation and evaluation skills in a supplied isolated-intersection scaffold. Implement a signal controller that changes phase or green allocation in response to observations.

A bounded design question

Example: Can a detector-based actuated or queue-responsive controller reduce total intersection delay relative to a justified fixed-time plan when demand changes? Use a small intersection with a simple documented phase plan. A network, complex turning movements and detailed lane changing are not required.

Required implementation

- Represent approach arrivals and queues, stop-line discharge and phase-dependent movement permission. Reuse car-following dynamics where practical, or justify a simpler queue model and check its discharge behavior. The scaffold supplies geometry and basic intersection mechanics, not the student's final control logic.
- Specify compatible movements, minimum/maximum green, intergreen or clearance assumptions, and service constraints. Include pedestrian service if pedestrians are modeled; otherwise state the scope and avoid claiming pedestrian benefits.
- Use detector or estimated-state inputs through the shared observation interface. Record the phase decision, reason, applied timing and response. Enforce the phase-transition constraints and a fallback plan when observations fail.

Minimum evaluation

- Compare the selected actuated or adaptive strategy with a justified fixed-time baseline using the same demand, initial queues and analysis horizon. Avoid selecting an artificially poor baseline.
- Test nominal balanced demand, a directional demand shift and a detector fault or delayed measurement. Include a heavy-demand case that exposes storage, residual queues or unmet demand.
- Report total and approach-specific delay, throughput, maximum/residual queues and service imbalance. Confirm that no incompatible movements receive green together in the modeled phase plan. Explain any assumptions behind surrogate safety measures.

Literature roles and presentation

Use at least two substantive sources for the final pathway, including a controller method or authoritative signal-control reference. One partner may own phase and controller logic; the other owns observation integration, baseline timing and evaluation. Both must explain a phase transition and the resulting queue response.

Present an interpretable phase timeline alongside queues and delay, defend the baseline, and show a case where the design needs revision. CEE281 adds a controller comparison or a sensitivity/ablation study. Incorporating connected-vehicle data is optional; it does not require completing the separate CAV pathway.

Design reviews and assessment requirements

Technical assessment principle	Scoring
Implementation and appropriate scope	T rubric
Verification and independent evaluation	T rubric
Fair experiments and interpretation	T rubric
Literature and reproducibility	T rubric
Named individual contribution	I rubric

There are no midterm or final examinations. Technical submissions account for 70% of the course grade: intermediate reviews 1, 2 and 3 carry 10%, 15% and 20%, and the final technical submission carries 25%. Individual progress presentations carry 15%; the final presentation and project questions carry 15%. Each technical score is 60% shared evidence and 40% individual work. Use the checkpoint-specific rubrics in the Course Grading Guide; the principles above describe their common emphasis.

Individual progress presentation criterion	Points
Explains the model and assumptions	30
Connects actual code to results	30
Diagnoses a problem and proposes a useful test	25
Responds to feedback and identifies next steps	15

Update one design record: about two pages for the proposal, three to four for later reviews, and six for the final report, plus references, plots and code. Present decisions and evidence.

Common submission package

- README with one reproducible run command, dependencies, input definitions, expected outputs and known limitations.
- Versioned Python code and scenario configurations; plots with axes, units and scenario labels.
- Experiment record: prediction, change made, result, and interpretation.
- Source-to-design table identifying equations, parameter choices and reused code; concise AI-use and verification disclosure.
- Individual contribution record identifying the component, code location, tests and conclusions each student owns. Record design changes made in response to feedback.

Scope for individual projects

An individual student owns a coherent implementation and receives companion modules for tasks that would otherwise require a partner. Examples include implementing the dynamics with a supplied detector, using supplied video tracks to build an estimator, or integrating a supplied observation pipeline with the student's own controller. The scientific and explanatory standards remain the same; implementation breadth is reduced.

Intermediate submissions and final presentation

The default design has three intermediate submissions and a final presentation. Choose a preliminary CAV or signal pathway by Week 5 so reading and setup can start early. Confirm it by November 9 and demonstrate the prototype at intermediate review 3 on November 16. Both options are choices of required final work, not optional extra credit. The lecture sequence is unchanged; CAV students receive early reading and starter examples before the Week 10 lectures.

Formal review	Submission and presentation purpose
Intermediate 1 after October 14	Proposal, model sources, roles, interfaces and a minimum simulation example. Consider both final control pathways.
Intermediate 2 after October 28	Verified simulation and detectors, behavioral results, and preliminary CAV-or-signal choice with initial sources and scope.
Intermediate 3 after November 16	Observation/forecast results, component exchange, a functioning basic TMC, and confirmed final-pathway design with initial prototype.
Final after December 2	Executable CAV OR signal-control system and report. Present baseline comparisons, stress tests, limitations and individual contributions.

Flexible progress discussions

Use short progress slots as needed: show a result or failure, identify the relevant model and code, question each contributor, and agree the next test. Two or three slots may fit a typical discussion block.

Question sequence for any demonstration (examples)

- What operational or modeling question are you answering, and what did you predict?
- Which model and source support this implementation? Show the relevant equation and code.
- What check would reveal that this result is wrong?
- If I change this demand, parameter, sensor or interface, what should happen and why?
- What will you test next, and which possible result would change your conclusion?

Formal final presentation

Allow approximately 12 minutes per pair for the final presentation: four minutes of demonstration and four minutes of explanation and project questions per student. An individual project uses eight minutes.